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McKinsey Sustainability

Advancing adaptation: Mapping hazards, exposure, and costs

Presentation document

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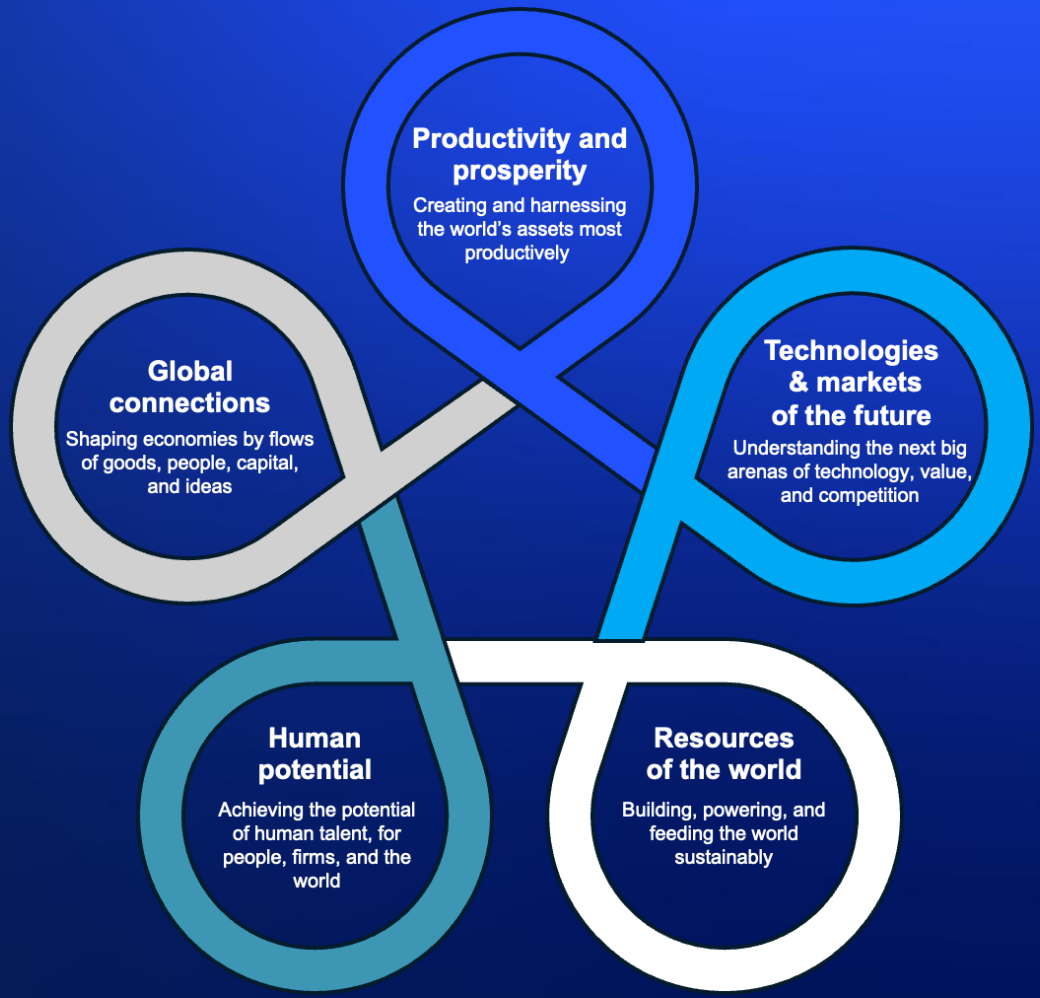
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Sustainability is a journey with a complex roadmap.

To move from plans to progress and from hype to results requires ambition, innovation, and transformation experience.

We help companies quickly scale what works, fuel growth while cutting carbon, and turn sustainability into a competitive edge and source of resilience



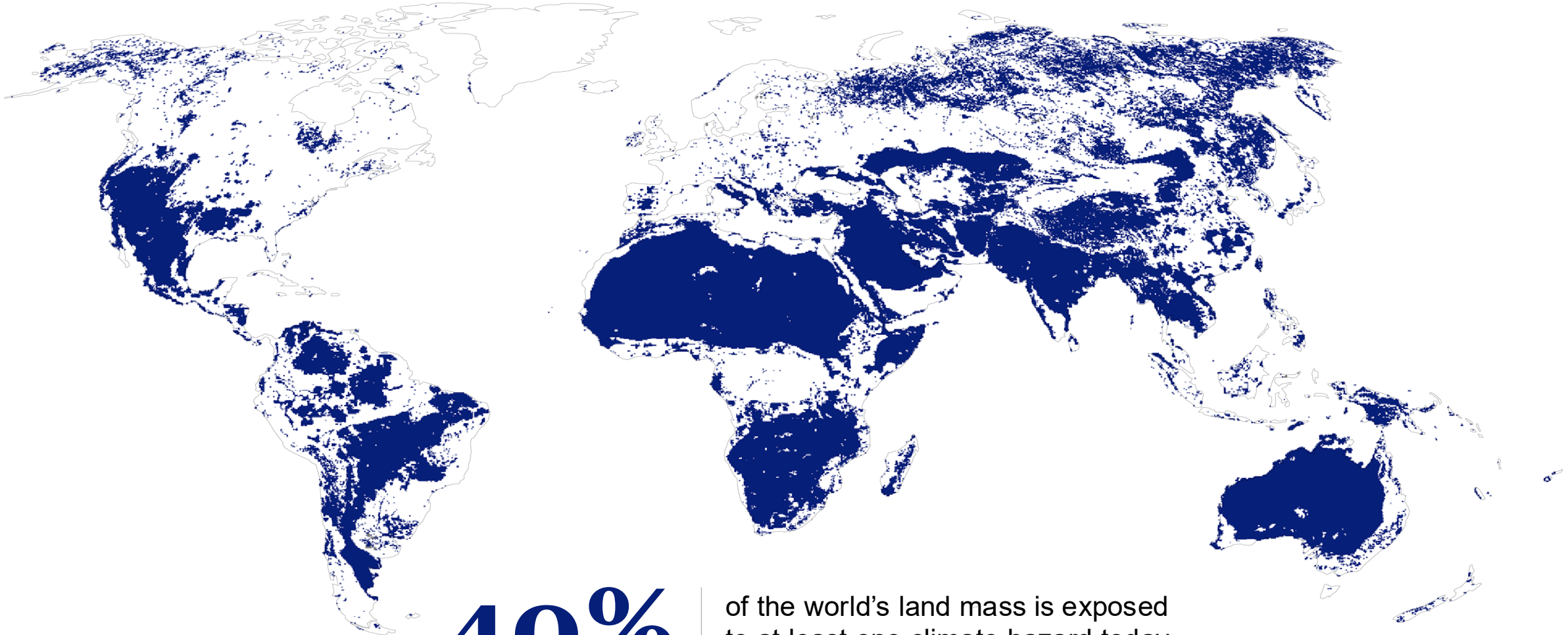


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Today the world already experiences a range of climate hazards

Places exposed to heat, wildfire, drought, or flooding, as defined by developed-economy protection standards, today[1]



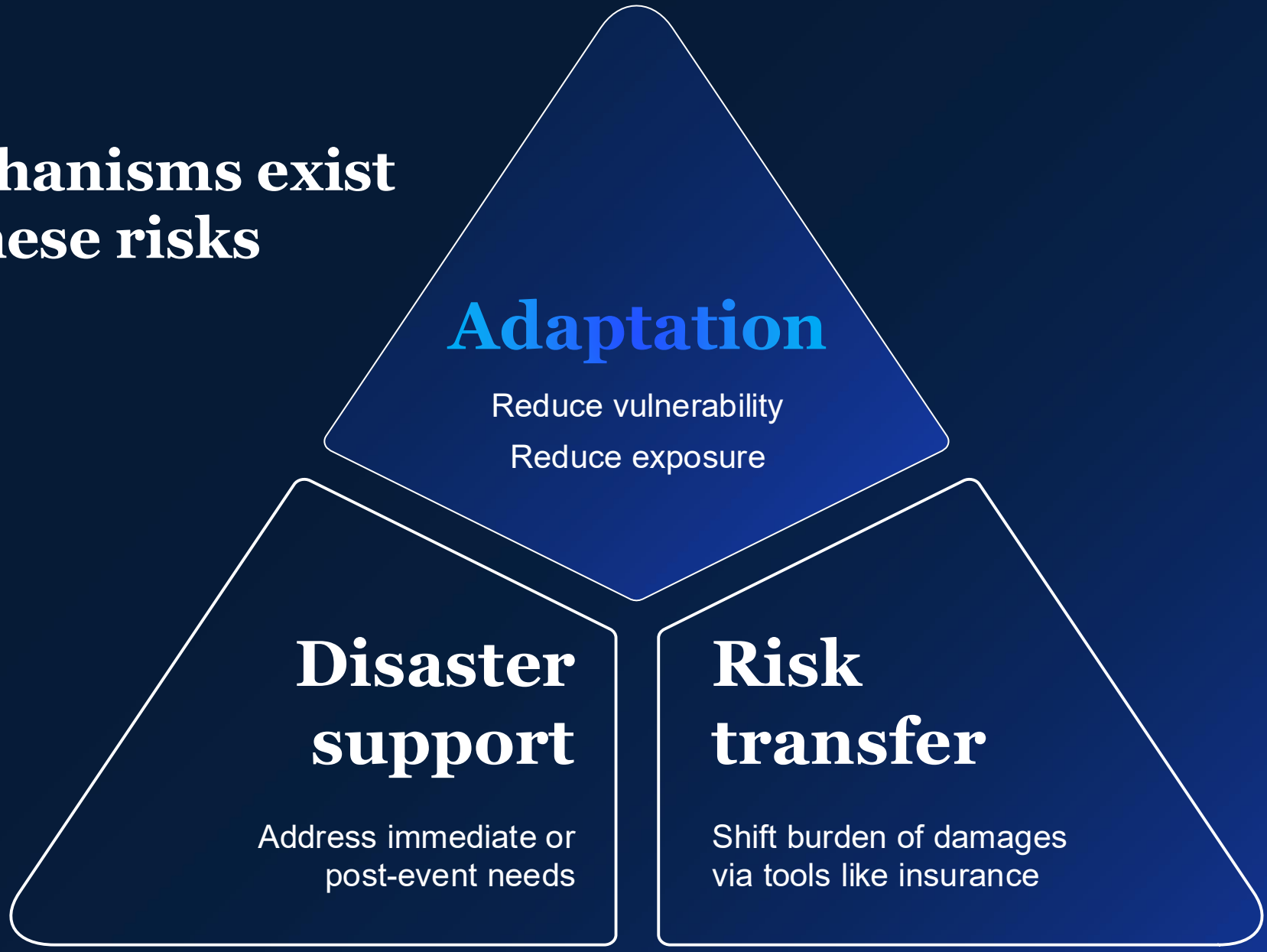
40% of the world's land mass is exposed to at least one climate hazard today

1. Observations of global temperatures indicate that the Earth warmed by 1.1°C relative to pre-industrial levels in the decade prior to 2020, which is when our analysis starts and what we refer to as "today."

Source: Advancing Adaptation: Mapping costs from cooling to coastal defenses, McKinsey Global Institute, December 2025

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Various mechanisms exist to manage these risks



Source: Advancing Adaptation: Mapping costs from cooling to coastal defenses, McKinsey Global Institute, December 2025

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There are many proven adaptation solutions

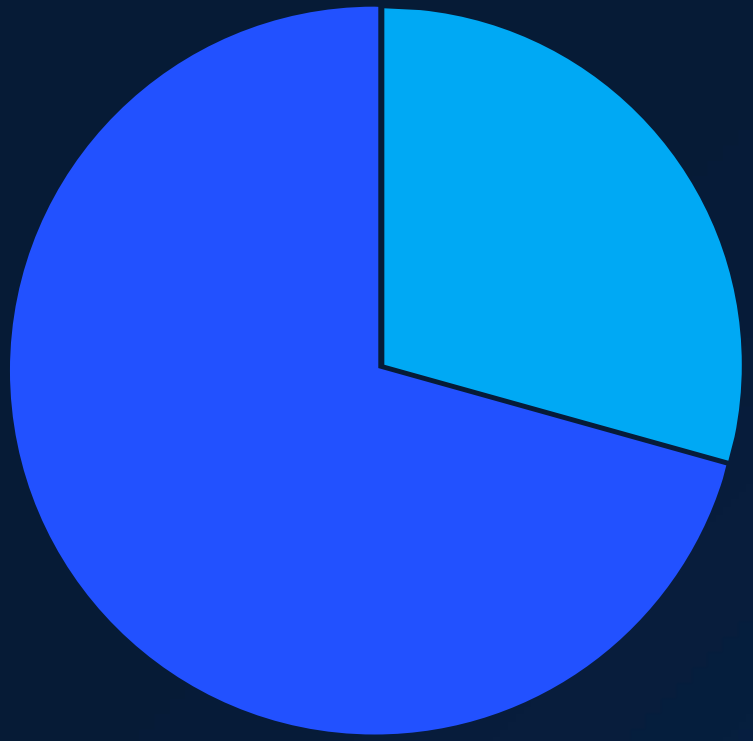
XX Solutions examined in this research

	Heat	Wildfire	Drought	Flooding	Multi-hazard
Reduce vulnerability	Active cooling (e.g., Air conditioning, fans) Cooling shelters for outdoor workers and vulnerable populations Personal cooling Urban trees Passive building cooling White roofs District cooling Cold-chain food transportation & storage	Undergrounding power lines Fuel management Retrofitting buildings with fire-resistant materials and better air filtration Defensible space guidelines Utility vegetation management & pole hardening	Irrigation Crop shade cover Drought-resistant seeds and livestock Seed priming Fertilizer application technology	Detention basins Stormwater networks Levees Swales Floodproofing Sea dikes Mangroves Dune and wetland restoration Permeable surfaces	Early warning systems Backup IT, power and water for critical systems Buffers for weather disruptions Diversification of production regions and logistics routes Supplier diversification
Reduce exposure	Working hours adjustments for outdoor workers	Seasonal closures / access management Land zoning in high wildfire risk locations	Herd management (early sales/preventative destocking)	Land zoning in high flood risk locations Managed retreat from at-risk coasts	Supply chain relocation

Source: Advancing Adaptation: Mapping costs from cooling to coastal defenses, McKinsey Global Institute, December 2025

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4.1B | people are currently exposed to heat, wildfire, flooding, and drought



1.2B
Protected

64%
live in high- and middle-income places

2.9B
Limited protection

77%
live in low-income places

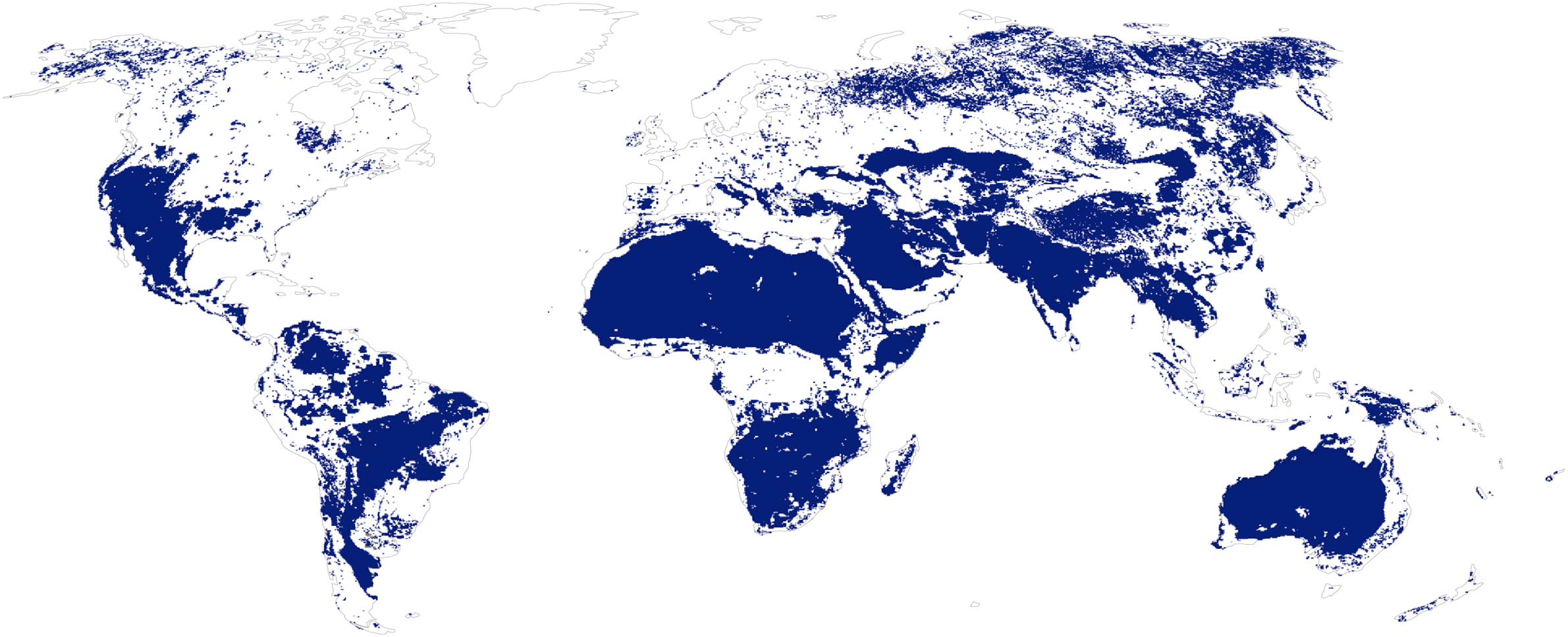
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Climate hazards will evolve: but not everywhere, not in the same way, and not all at once

Places exposed to heat, wildfire, drought, or flooding as defined by developed-economy protection standards

Today[1] | At 1.5°C | At 2.0°C



1. Observations of global temperatures indicate that the Earth warmed by 1.1°C relative to pre-industrial levels in the decade prior to 2020, which is when our analysis starts and what we refer to as "today."

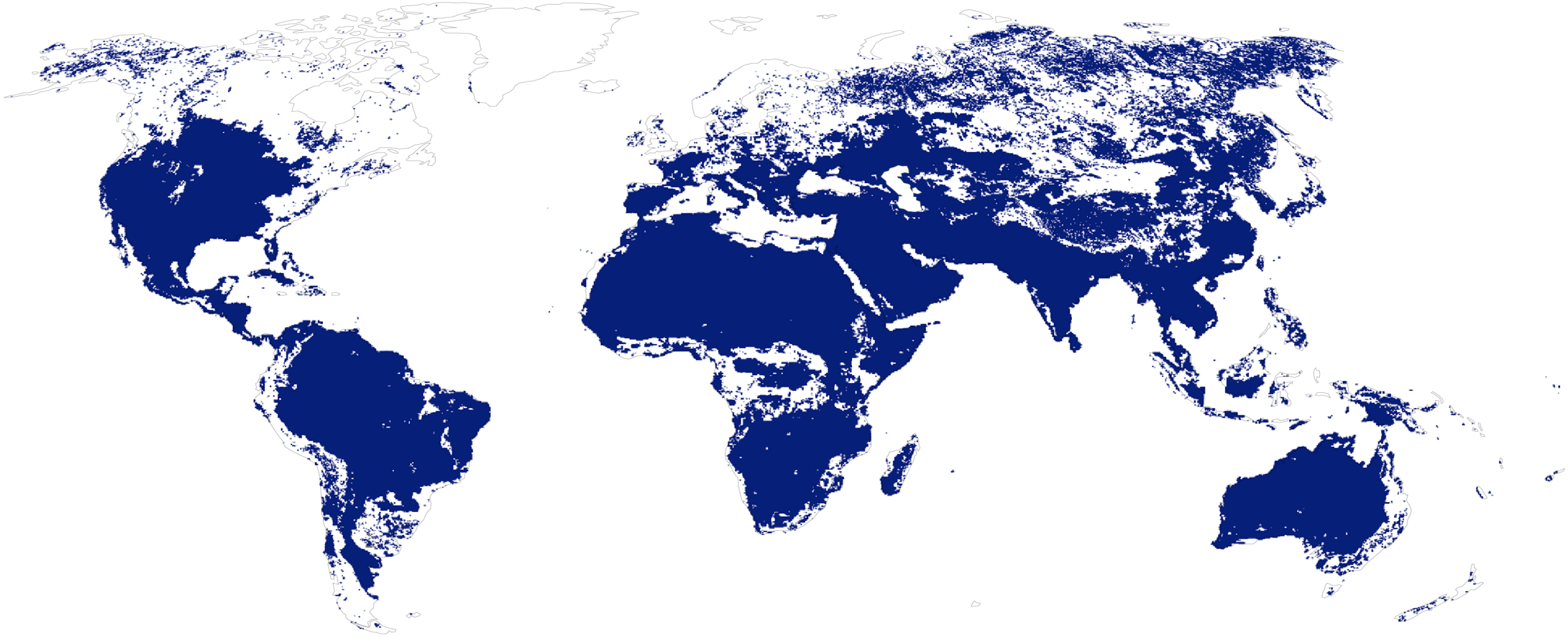
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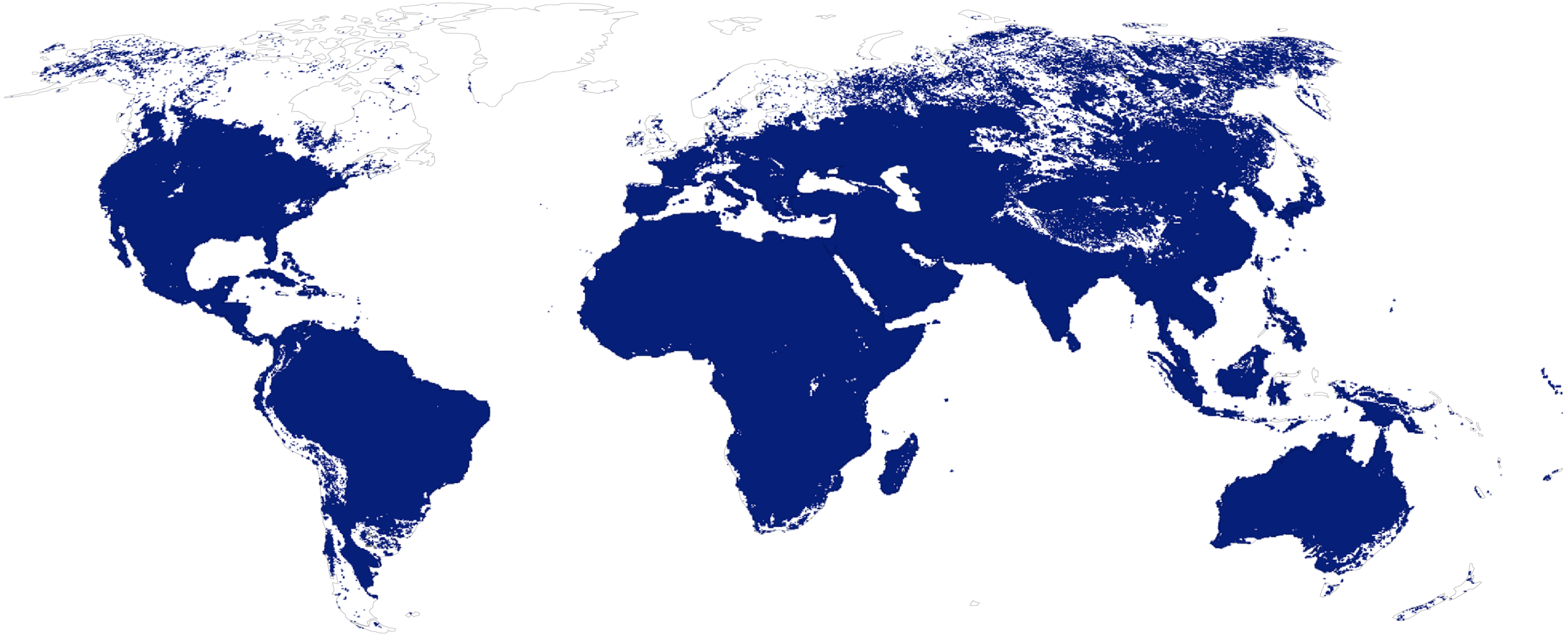
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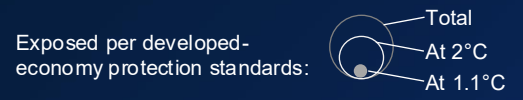
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Across regions, exposure to heat and drought would increase most

Share of population in places exposed to hazards, by region, 2050, *% of total*



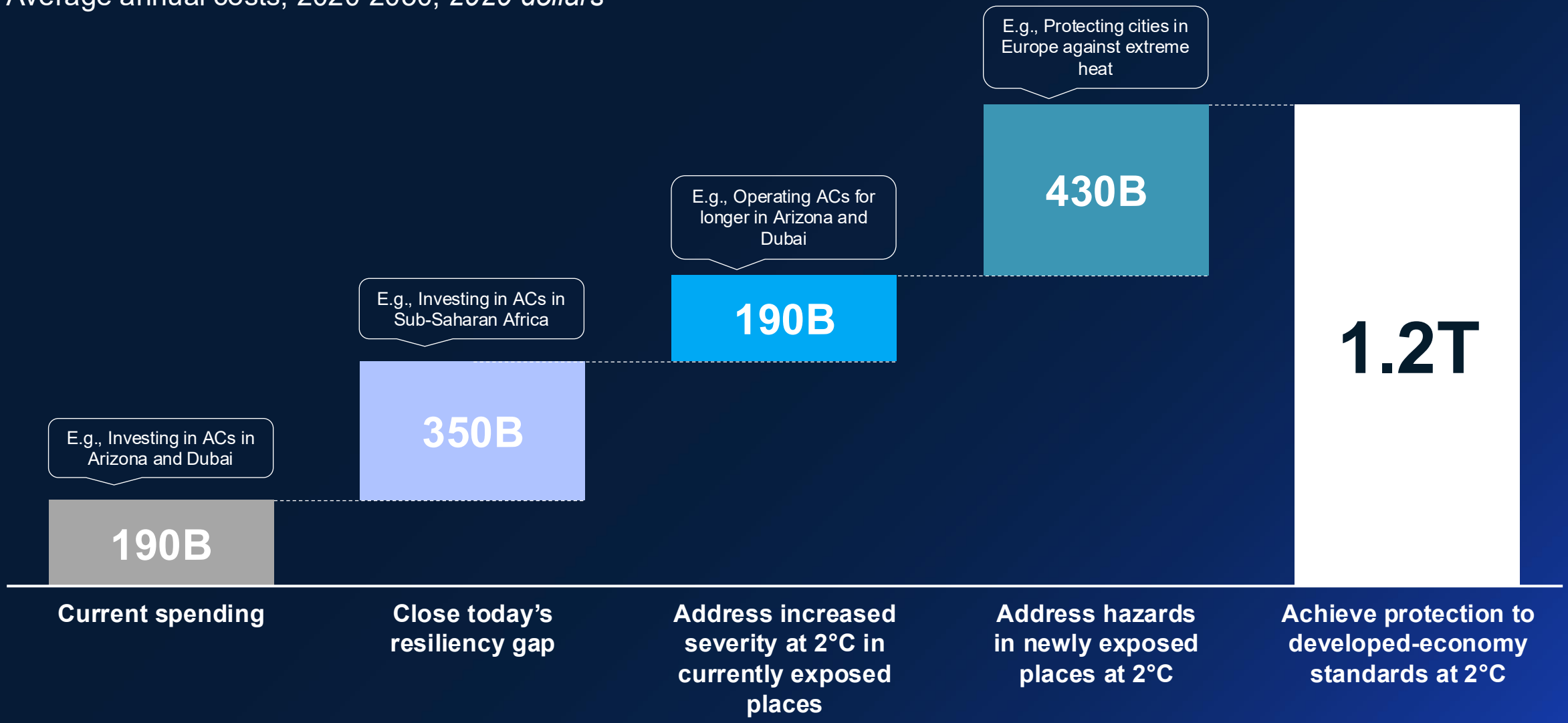
Region	Heat stress	Heatwaves	Nonsurvivable heat	Drought[2]	Wildfire weather	Excess rainfall flooding	Riverine flooding	Coastal flooding
Sub-Saharan Africa								
India								
Emerging Asia								
Greater China								
Middle East & North Africa								
Latin America & Caribbean								
Other Europe & Central Asia								
North America								
EU30[3]								
Advanced Asia								

Source: Advancing Adaptation: Mapping costs from cooling to coastal defenses, McKinsey Global Institute, December 2025

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Protecting to developed-economy standards at 2°C could cost $1.2 trillion annually by 2050

Average annual costs, 2020-2050, *2020 dollars*



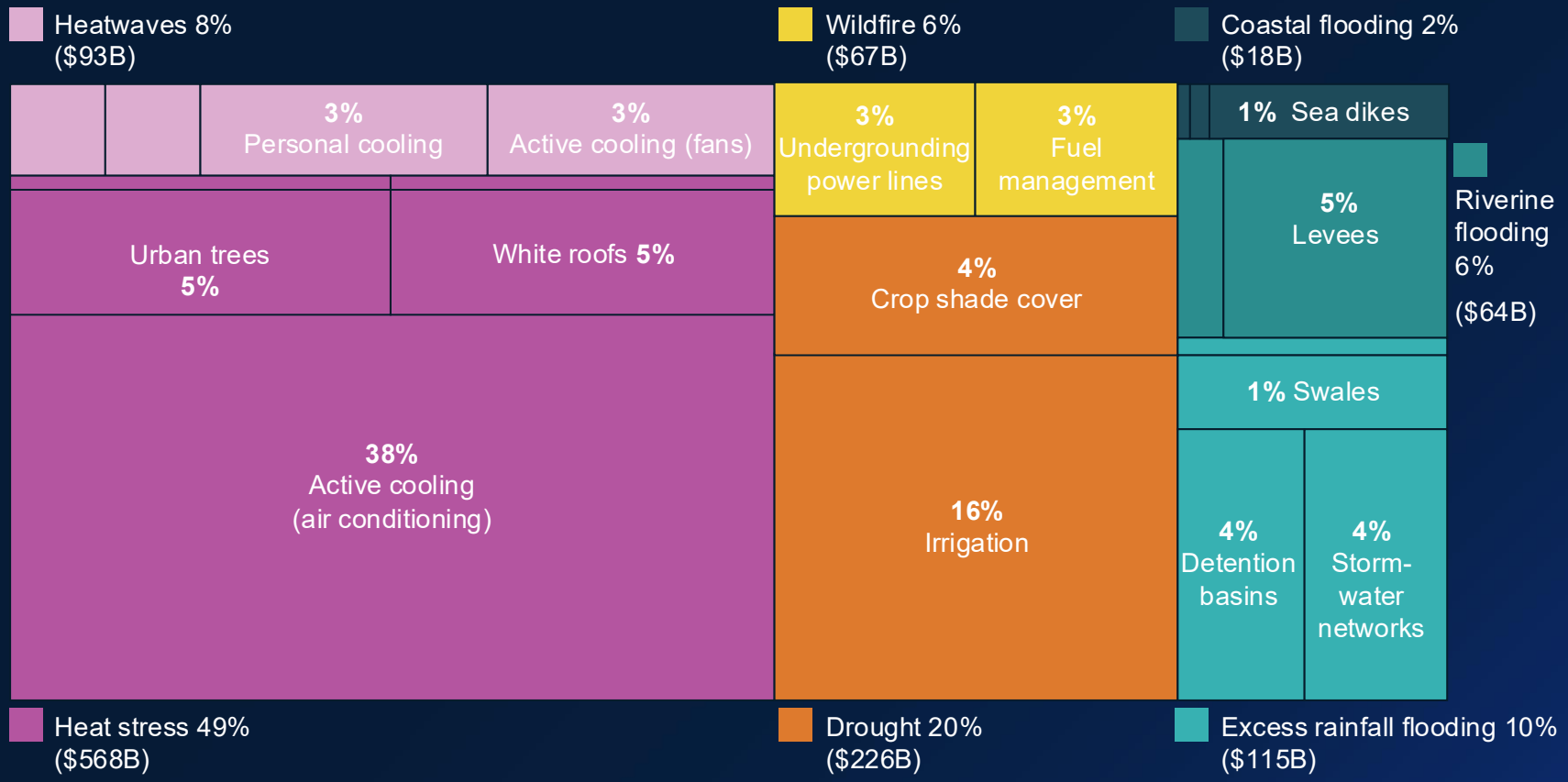
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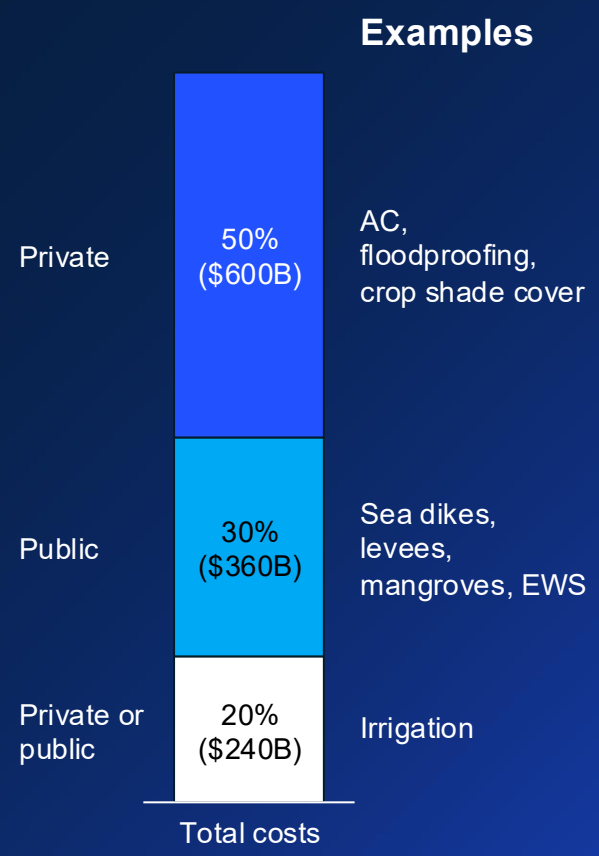
Air conditioning and irrigation account for >50% of costs

Distribution of annual costs to adapt to developed-economy standards at 2ºC, 2020-2050, *% of total, $ billion (2020 dollars)*

By hazard



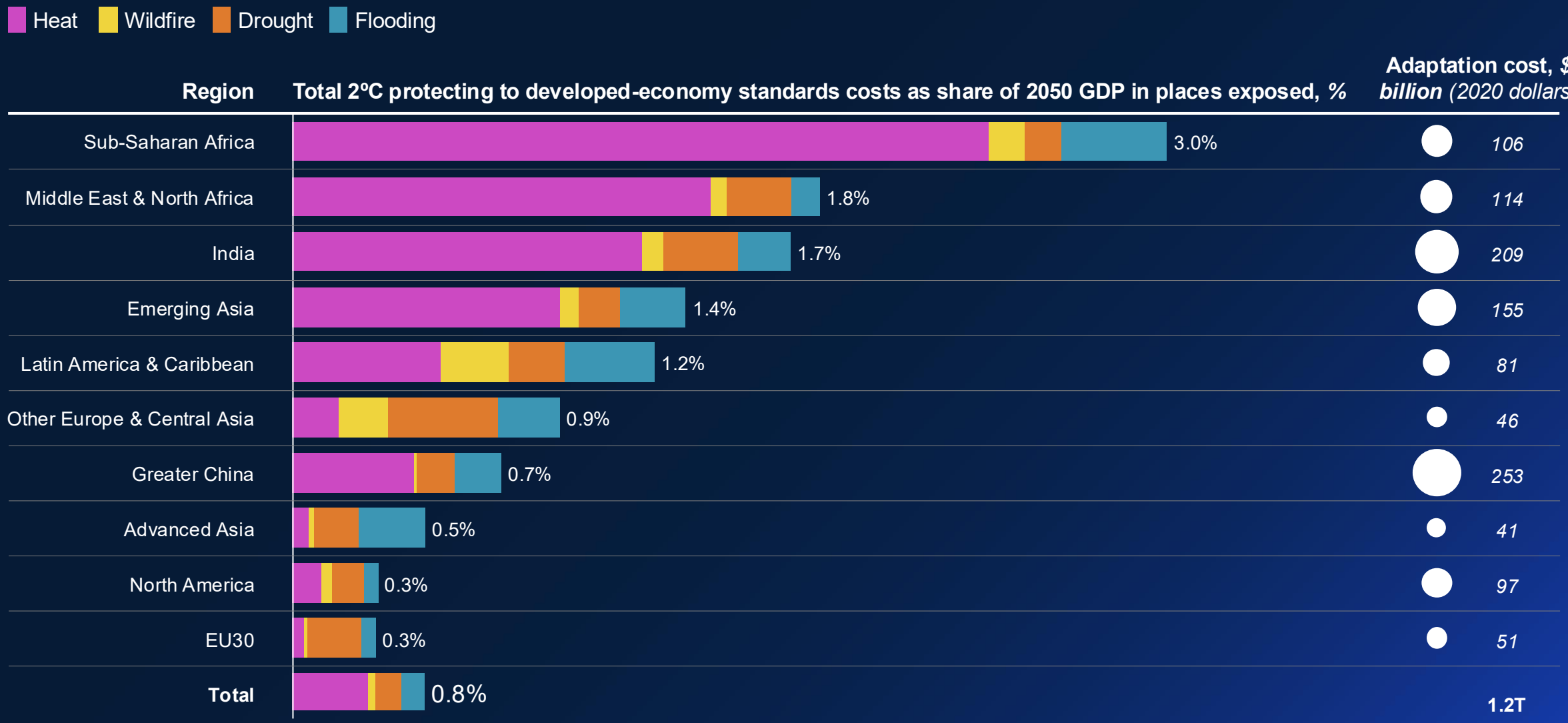
By stakeholder implementation



Source: Advancing Adaptation: Mapping costs from cooling to coastal defenses, McKinsey Global Institute, December 2025

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Sub-Saharan Africa, MENA, and India have the highest adaptation costs as a share of GDP



Source: Advancing Adaptation: Mapping costs from cooling to coastal defenses, McKinsey Global Institute, December 2025

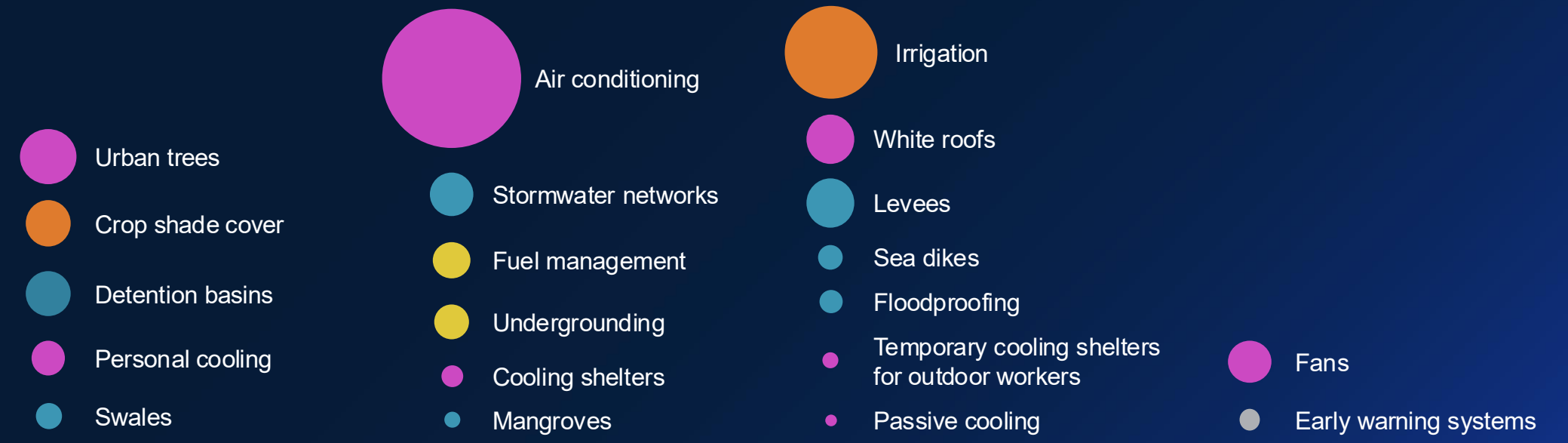
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Adaptation is a good buy with BCRs of all measures > 1.5

Hazard: Heat · Wildfire weather · Drought · Flooding

Size of bubble represents annual operating and amortized capital costs to adapt to hazards to developed-economy standards at 2°C, 2020-2050, *$ billion (2020 dollars)*

$100B



Average BCR of 20 costed measures increases from 3 today to 7 at 2°C of warming

Source: Advancing Adaptation: Mapping costs from cooling to coastal defenses, McKinsey Global Institute, December 2025

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Why stakeholders might not pay all the $1.2T

Capacity to pay

Protecting to developed-economy standards costs about $130 per person today, but current spend is only $45 per person; competing spending priorities also exist

Alignment of incentives

Funders of adaptation measures may not be the direct beneficiaries of protection provided

Nature of adaptation benefits

Benefits materialize only if hazards occur, which weakens incentives to invest

Risk awareness

Areas that have not experienced hazards recently underestimate risks, e.g., funding spikes after events (like LA wildfire), then fade

Risk appetite

Netherlands designs coastal barriers for 1-in-10,000-year events — vs. 100-year norm

Execution challenges

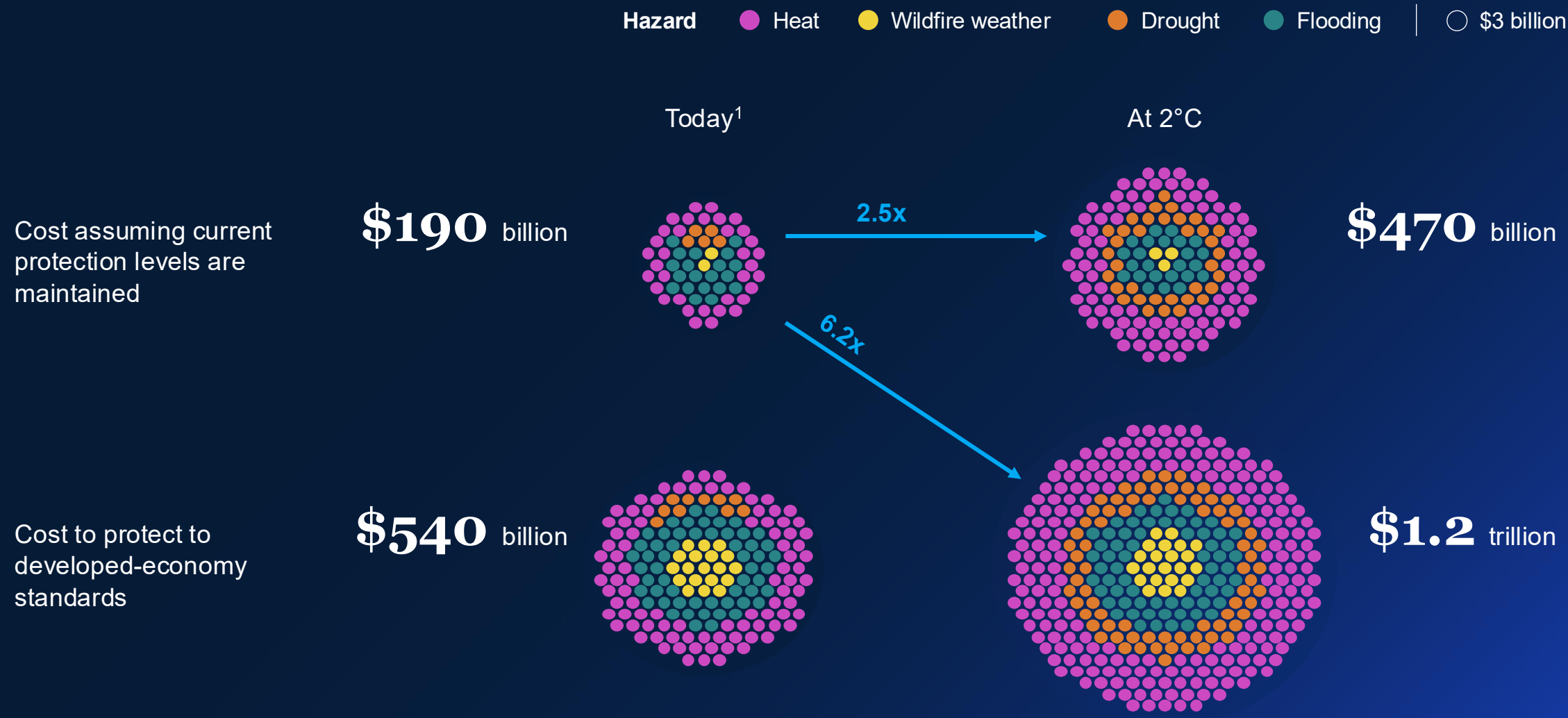
Supply chain, political will, technical, coordination, and other hurdles — e.g., decade-long talks for Netherlands' Room for the River program

Source: Advancing Adaptation: Mapping costs from cooling to coastal defenses, McKinsey Global Institute, December 2025

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But how much might get spent?

Average annual costs to adapt to hazards, 2020-2050, *2020 dollars*



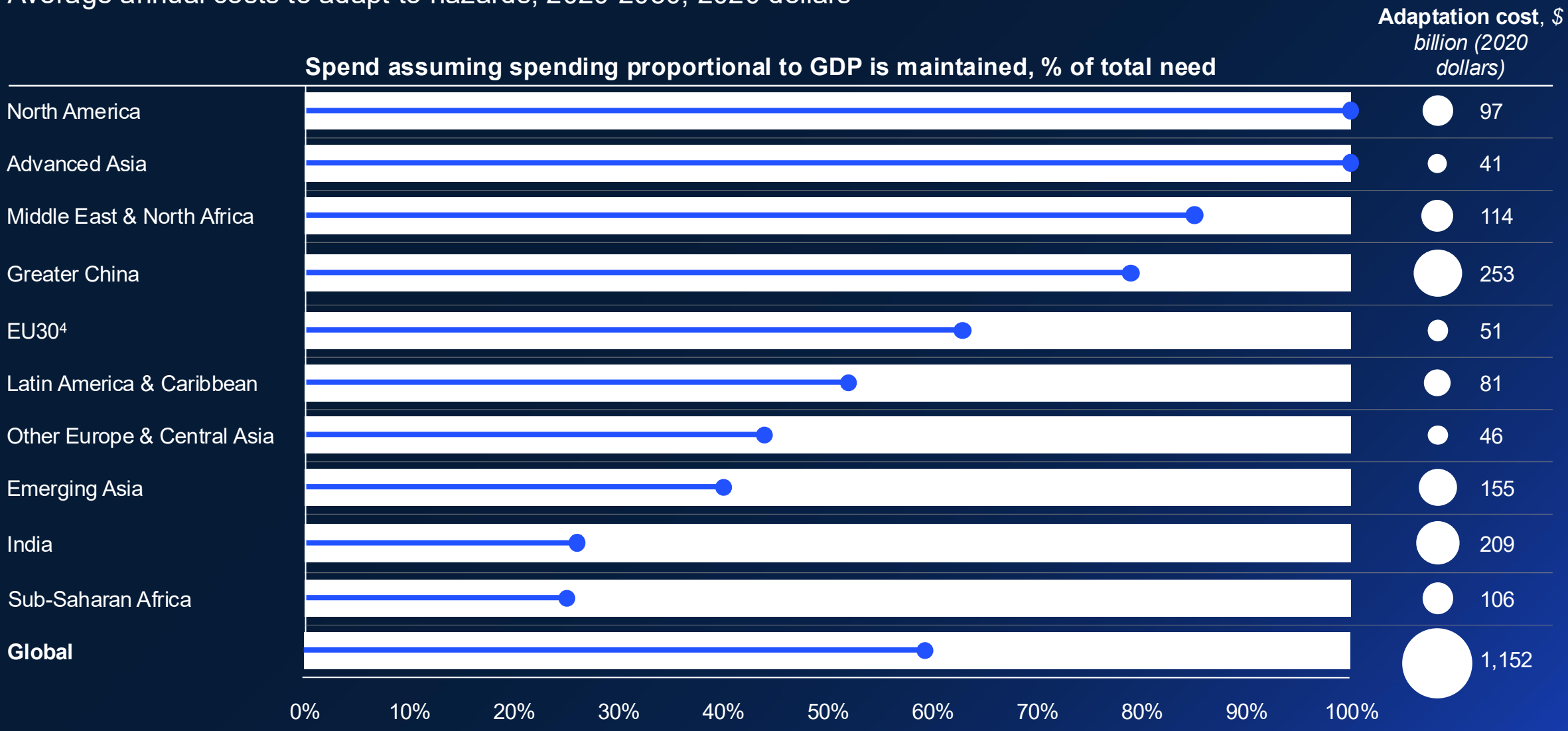
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Maintaining spending proportional to GDP could enable 60% of global adaptation needs to be met

Average annual costs to adapt to hazards, 2020-2050, 2020 dollars



Source: Advancing Adaptation: Mapping costs from cooling to coastal defenses, McKinsey Global Institute, December 2025

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What role can different stakeholders play?

Individuals

Protect against heat (e.g., AC, passive cooling)

Potentially address other hazards (e.g., floodproofing)

Companies

Capture opportunities to provide products/services

Adapt operations (e.g., harden sites, support workers, build resilient supply chains)

Bake adaptation into capex planning



Governments

Support large-scale infra projects (e.g., sea dikes, levees, district cooling)

Design for non-stationarity & uncertainty

Provide social protection and set standards

Financial institutions

Expanding finance for the broader adaptation ecosystem

Facilitating **capital flows** and developing **institutional partnerships**

Capturing value from **underwriting innovation**

Picking up the pieces

Adapt proactively or bear the risk

Absorb residual costs when protection is absent or insufficient (e.g., taxpayers, insurers)

Source: Advancing Adaptation: Mapping costs from cooling to coastal defenses, McKinsey Global Institute, December 2025

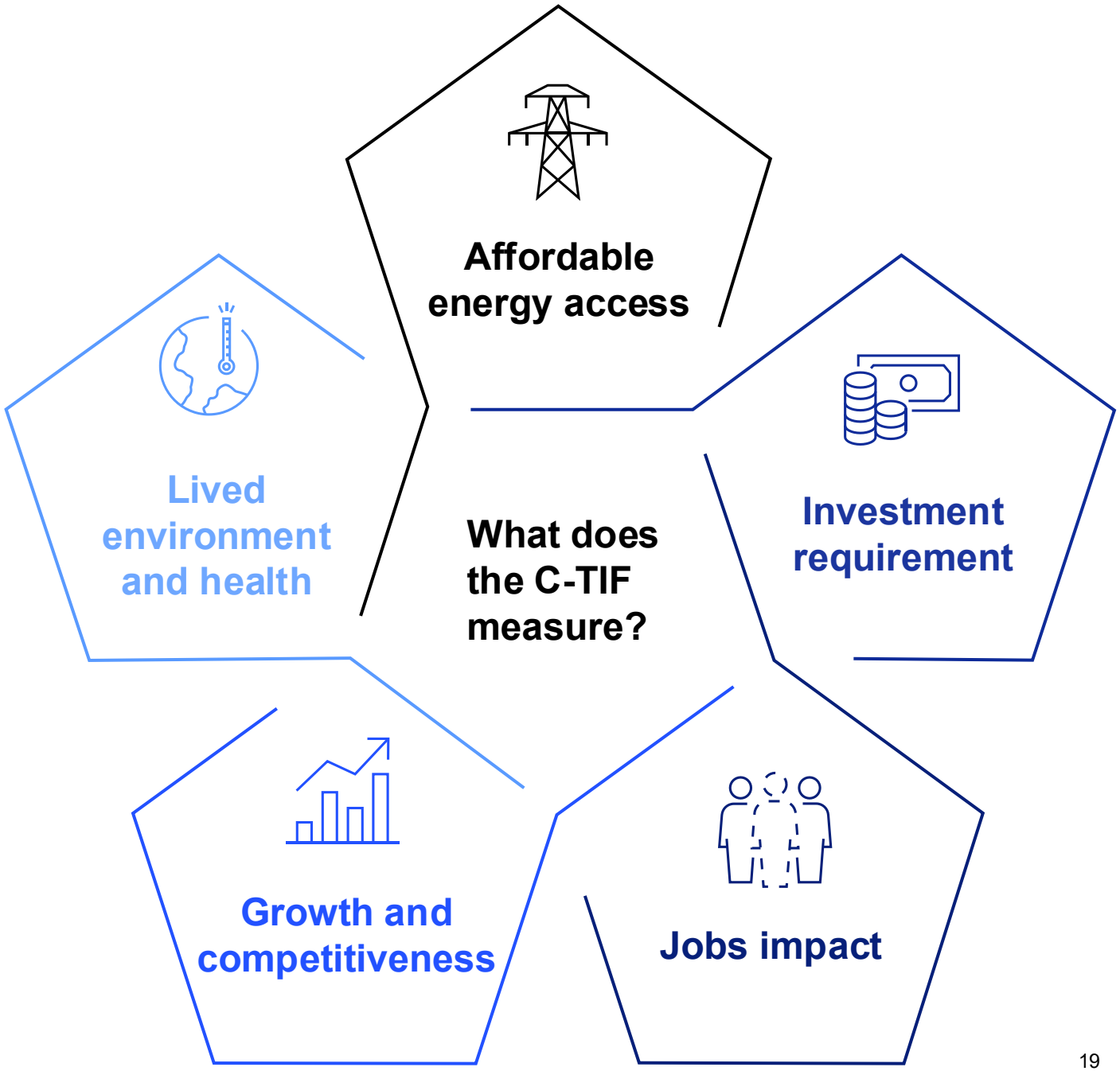
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Climate Transition Impact Framework (C-TIF)

A structured forward-looking approach that quantifies the socioeconomic impacts of climate action

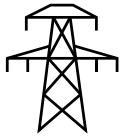
Developed together with:

UN trade & development · NGFS · McKinsey Sustainability



The C-TIF is comprised of five dimensions, that together tell a comprehensive story of socioeconomic impact

Affordable energy access



While nearly all populations in most regions have access to electricity, **685 million people remain off grid,** of whom more than **80% are in sub-Saharan Africa.** This dimension includes metrics on **electricity and energy costs,** and **energy access,** which are all likely to be significantly impacted by shifts in energy supply

Lived environment & health



The implications of the climate transition on the lived environment and human health will **vary by country and even community**. This dimension includes metrics on **air pollution,** changes in **forest coverage**, and **climate hazard** exposure which vary by country

Growth & competitiveness



The impact of the net-zero transition on economic growth is a **complex and nuanced.** Many interplaying factors could influence whether transitions **support or constrain income growth,** and when these effects are likely to take place. Understanding these dynamics can be crucial to decision makers ensuring **buy in on transition plans from businesses and people**

Jobs impact



The transition towards a net-zero economy will drive **substantial shifts in labour markets**, as jobs in conventional sectors are displaced by opportunities in the deployment, operation, and maintenance of **low-carbon systems and technologies**. In some hydrocarbon dependent economies, these effects could be **particularly pronounced**

Investment requirement

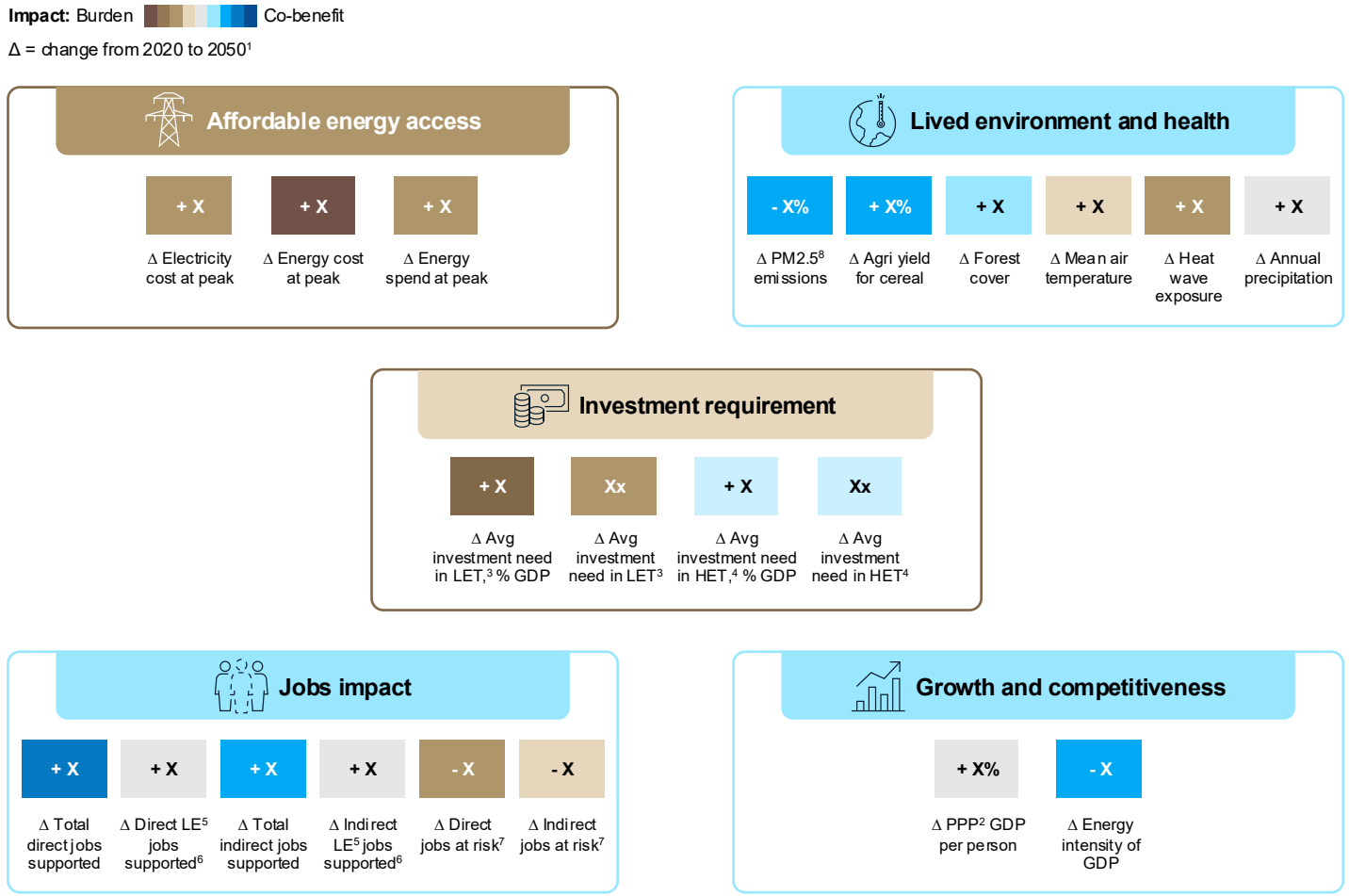


Investment patterns **will need to shift dramatically in all countries** to achieve the deployment of low-carbon technologies required by the transition. Rather than speak to where finances should come from, this dimension seeks to detail **where investments should flow to**

C-TIF results summary table

- To understand socioeconomic impacts, one can **classify the dimensions as presenting potential co-benefits & burdens**
- Socioeconomic **co-benefits can be used to unlock multi-dimensional value** allowing for more robust implementation pathways
- Identifying **socioeconomic burdens can support risk mitigation** as mitigation measures can be part of initial planning

Outputs are illustrative and not a prediction or projection. They are based on scenarios, assumptions and data used and are intended to illustrate how the C-TIF framework might be applied. Decision-makers should understand the limitations and uncertainties inherent in the underlying modelling.



1. Except for the energy cost subdimension and the mean air temperature metric 2.Purchasing Power Parity 3. Low Emissions Technology 4. High Emissions Technology 5. Low emissions 6. from total jobs supported 7. The "-" polar sign for jobs at risks indicates an increase. 8. Particulate Matter

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